THE LARGEST OPEN WHERE A REPRESENTABLE MORPHISM HAS PROPERTY P DESCENDS TO STACKS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2299-2327 of sources/stacks-stacks-properties/pinned-original-source.tex; proof span is lines 2329-2363, the complete author proof (33 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 4 of queue_stacks-properties.json (pb=1896, stmt 2299-2327, proof 2329-2363).

Editorial note. stacks-properties wave-1: fresh adjudication accept (2026-10-14); pool scanned while CONTROL.md is down (0 bytes) - §2 row deferred per Chief, claim reported to Chief 08:0x.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma lemma-check-property-covering: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 683–693. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-immersion-into-presentation: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1896–1910. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{P}, \mathcal{Q}, \mathcal{R}$ be properties of morphisms of algebraic spaces. Assume*

(1) *$\mathcal{P}, \mathcal{Q}, \mathcal{R}$ are fppf local on the target and stable under arbitrary base change,*
(2) *smooth $\Rightarrow \mathcal{R}$,*
(3) *for any morphism $f : X \to Y$ which has $\mathcal{Q}$ there exists a largest open subspace $W(\mathcal{P}, f) \subset X$ such that $f|_{W(\mathcal{P},f)}$ has $\mathcal{P}$, and*
(4) *for any morphism $f : X \to Y$ which has $\mathcal{Q}$, and any morphism $Y' \to Y$ which has $\mathcal{R}$ we have $Y' \times_Y W(\mathcal{P}, f) = W(\mathcal{P}, f')$, where $f' : X_{Y'} \to Y'$ is the base change of f.*

Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks representable by algebraic spaces. Assume f has $\mathcal{Q}$. Then

(A) *there exists a largest open substack $\mathcal{X}' \subset \mathcal{X}$ such that $f|_{\mathcal{X}'}$ has $\mathcal{P}$, and*

Date: stacks-project:stacks-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(B) *if $\mathcal{Z} \to \mathcal{Y}$ is a morphism of algebraic stacks representable by algebraic spaces which has $\mathcal{R}$ then $\mathcal{Z} \times_{\mathcal{Y}} \mathcal{X}'$ is the largest open substack of $\mathcal{Z} \times_{\mathcal{Y}} \mathcal{X}$ over which the base change $id_{\mathcal{Z}} \times f$ has property $\mathcal{P}$.*

Proof. Choose a scheme V and a surjective smooth morphism $V \to \mathcal{Y}$. Set $U = V \times_{\mathcal{Y}} \mathcal{X}$ and let $f' : U \to V$ be the base change of f. The morphism of algebraic spaces $f' : U \to V$ has property $\mathcal{Q}$. Thus we obtain the open $W(\mathcal{P}, f') \subset U$ by assumption (3). Note that $U \times_{\mathcal{X}} U = (V \times_{\mathcal{Y}} V) \times_{\mathcal{Y}} \mathcal{X}$ hence the morphism $f'' : U \times_{\mathcal{X}} U \to V \times_{\mathcal{Y}} V$ is the base change of f via either projection $V \times_{\mathcal{Y}} V \to V$. By our choice of V these projections are smooth, hence have property $\mathcal{R}$ by (2). Thus by (4) we see that the inverse images of $W(\mathcal{P}, f')$ under the two projections $\mathrm{pr}_i : U \times_{\mathcal{X}} U \to U$ agree. In other words, $W(\mathcal{P}, f')$ is an R-invariant subspace of U (where $R = U \times_{\mathcal{X}} U$). Let $\mathcal{X}'$ be the open substack of $\mathcal{X}$ corresponding to $W(\mathcal{P}, f)$ via Lemma lemma-immersion-into-presentation. By construction $W(\mathcal{P}, f') = \mathcal{X}' \times_{\mathcal{Y}} V$ hence $f|_{\mathcal{X}'}$ has property $\mathcal{P}$ by Lemma lemma-check-property-covering. Also, $\mathcal{X}'$ is the largest open substack such that $f|_{\mathcal{X}'}$ has $\mathcal{P}$ as the same maximality holds for $W(\mathcal{P}, f)$. This proves (A).

Finally, if $\mathcal{Z} \to \mathcal{Y}$ is a morphism of algebraic stacks representable by algebraic spaces which has $\mathcal{R}$ then we set $T = V \times_{\mathcal{Y}} \mathcal{Z}$ and we see that $T \to V$ is a morphism of algebraic spaces having property $\mathcal{R}$. Set $f'_T : T \times_V U \to T$ the base change of f'. By (4) again we see that $W(\mathcal{P}, f'_T)$ is the inverse image of $W(\mathcal{P}, f)$ in $T \times_V U$. This implies (B); some details omitted. □